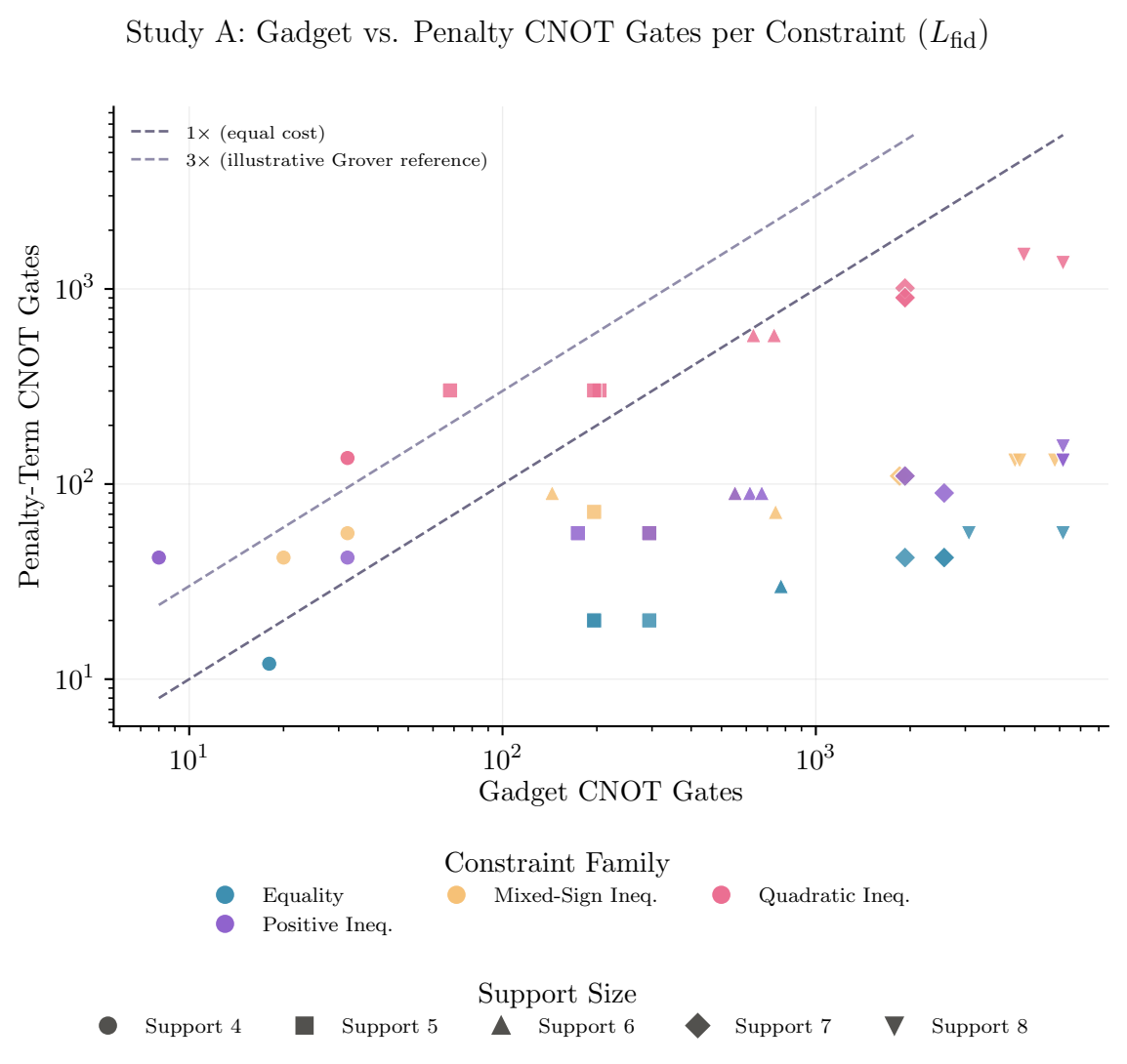

Study A: Gadget vs. Penalty CNOT Gates per Constraint (L_{fid})
1× (equal cost)
3× (illustrative Grover reference)
Penalty-Term CNOT Gates
10^1
10^2
10^3
Gadget CNOT Gates
10^1
10^2
10^3
Constraint Family
Equality
Mixed-Sign Ineq.
Quadratic Ineq.
Positive Ineq.
Support Size
Support 4
Support 5
Support 6
Support 7
Support 8